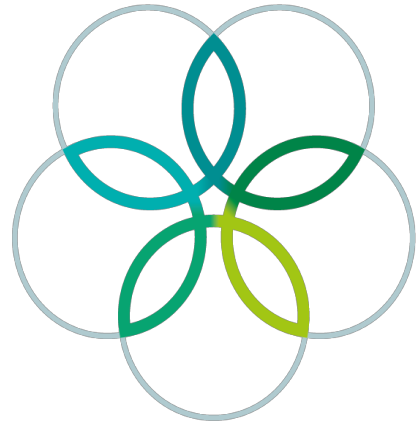
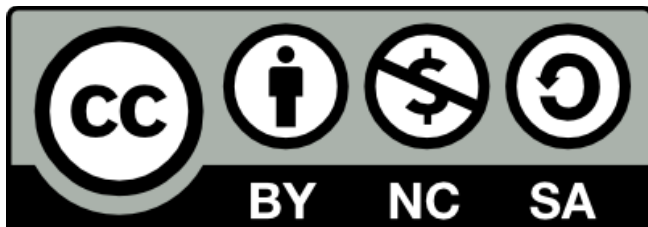


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IVth International Biology Olympiad (Utrecht, the Netherlands, 1993)

Theoretical Test

The theoretical test consists of two parts, A and B.

Part A consists of 111 multiple choice questions.

You have got two answering sheets for this test A:

sheet 1: questions 1 - 100

sheet 2: questions 101 - 111.

Use a pencil to fill out the multiple choice questions on the answering sheets.

Part B consists of 29 open questions.

For this part B you have to fill out your answers in the given space on the test itself.

For each question in part A and B the obtainable scores are indicated in brackets. The total score is:

Totally:	Part A:	136 points
	Part B:	70 point
	Together:	206 points

The figures and graphs are in an appendix and will be handed out to you separately. You will need these figures and graphs for both part A and B, so please keep them during the whole contest.

You have three and a half hours to complete the test.

How you divide the available time is up to you, but you are advised not to spend more than two hours on part A.

Before starting, be sure that your name and participant number are correctly indicated on the two answering sheets of part A and on test B itself.

GOOD LUCK!

Theoretical test, Part A.

1.(2pt) As result of a mutation in a cell, a protein is formed with a different type of amino acid at the place of the amino acid lysine. In answering the question use the information given in the table with mRNA codons (fig. 1)

		second position				
		U	C	A	G	
first position (5' end)	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } Stop UAG }	UGU } Cys UGC } UGA } Stop UGG } Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ileu AUA } AUG } Met (start)	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Fig. 1

What amino acid is most likely to be built in, aspartic acid (Asp) or methionine (Met) and which nucleotide is mutated?

	amino acid	nucleotide mutated
A.	aspartic acid	adenine
B.	aspartic acid	thymine
C.	methionine	adenine
D.	methionine	thymine

2.(2pt) The table shows two different mRNA molecules and the two different protein molecules synthesized from them.

mRNA	protein
.....AGAGAGAGAGAGAGAGAGAGAGAG.....	P
.....AAUGAAUGAAUGAAUGAAUGAAUG.....	Q

How many different types of amino acids can be found in each protein molecule?

	P	Q
A.	1	3
B.	1	4
C.	2	3
D.	2	4

3.(1pt) A gene can be defined as:

- A. part of an RNA molecule, characterized by a certain order of nucleotides.
- B. a particular order of nucleotides in a DNA-molecule.
- C. part of a DNA-molecule, characterized by a certain order of nucleotides, which determines the synthesis of one protein.
- D. order of amino acids in a protein molecule.

.(1pt) Which one of the following items is not required for protein synthesis?

- A. mRNA
- B. ribosomes
- C. tRNA
- D. endoplasmic reticulum
- E. amino acids

5.(1pt) Which one of the following statements is true for enzymes?

- A. Enzymes lose some or all of their normal activity if their three dimensional structure is disrupted.
- B. Enzymes provide the activation energy necessary to initiate a reaction.
- C. The activity of enzymes is independent of temperature and pH.
- D. An enzyme acts only once and is then destroyed.

6.(1pt) A bacterial mRNA with a length of 360 nucleotides in length codes for a protein of:

- A. roughly 360 amino acids
- B. roughly 1080 amino acids
- C. exactly 120 amino acids
- D. less than 120 amino acids

7.(1pt) What is the most important function of the organelle P in figure 2?

- A. synthesis of proteins
- B. synthesis of ATP with energy from dissimilation reactions
- C. synthesis of ATP with energy from the sun
- D. transport of proteins

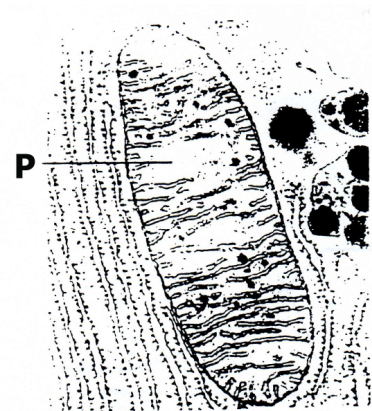


Fig. 2

8.(1pt) When a muscle cell has a shortage of oxygen will the pH decrease or increase? What substance is responsible for this change in pH?

- | | change in pH | caused by |
|----|--------------|-----------------------|
| A. | decrease | carbon dioxide |
| B. | decrease | lactate (lactic acid) |
| C. | increase | carbon dioxide |
| D. | increase | lactate (lactic acid) |

- 9.(1pt) An enzyme catalyses the reaction $P + Q \rightarrow R$.
The continuous line in the graph (Fig.3) shows the progress of this reaction in the absence of the enzyme. At time point t_1 , the enzyme that catalyses the reaction is added to the reaction mixture. ([Z] is representing the concentration of a compound Z).

Which line in figure 3 shows how the reaction will proceed ?

- A. A
- B. B
- C. C
- D. D
- E. E

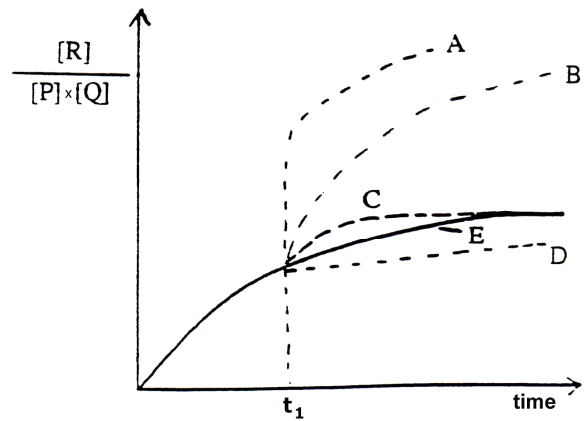
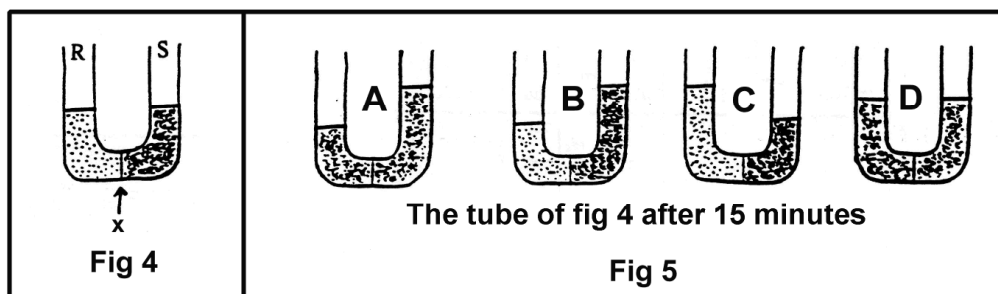


Fig. 3

- 10.(1pt) Which of the following processes causes the highest energy release?
- A. photolysis
 - B. glycolysis
 - C. Krebs cycle (citric acid cycle)
 - D. terminal oxidations
- 11.(1pt) What happens during glycolysis and what are the end products?
- A. Oxygen from CO_2 is replaced by hydrogen and glucose is formed.
 - B. As a result of splitting by fermentation of a glucose molecule, two molecules of tricarboxylic acid are formed and 2 molecules of ATP are synthesized.
 - C. As a result of the splitting of fat molecules, energy is released, which the cell needs.
 - D. Tricarboxylic acid is split up into carbon dioxide and water, as a result of which 36 molecules of ATP are synthesized.
- 12.(1pt) The concentration of an electrically neutral substance within a certain type of blood cell is much higher than it is in the surrounding blood plasma, yet the substance continues to move into the cell. The process by which this substance moves into the cell is called:
- A. osmosis
 - B. simple diffusion
 - C. facilitated diffusion
 - D. active transport
- 13.(2pt) An U-shaped tube is filled with two different solutions, R and S, separated by a semi-permeable membrane (X). R is hypotonic in relation to S, seen by the lighter tone in figure 4.



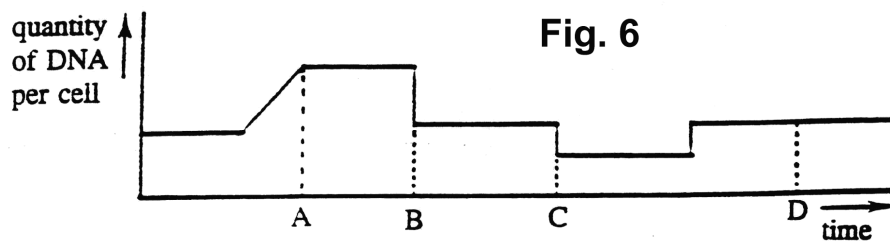
When an equilibrium is reached in the U-shaped tube of figure 4, are the surfaces equally high or not, and are the osmotic pressures equal or not (see also figure 5)?

- A. The right side is higher, the two solutions are isotonic
- B. The right side is higher, the right side is hypertonic
- C. The left side is higher, the right side is hypertonic
- D. The surfaces are equally high, the two solutions are isotonic

14.(1pt) Which one of the following pairs is correctly matched?

- A. chloroplast - storage of enzymes
- B. peroxisomes - cellular transportation
- C. nucleolus - site of ribosomal subunit synthesis
- D. lysosomes - power house of cell

15.(1pt) See figure 6. What letter in figure 6 indicates the end of meiosis I?



- A. A
- B. B
- C. C
- D. D

16.(1pt) Which organelles take part in the formation of a spindle?

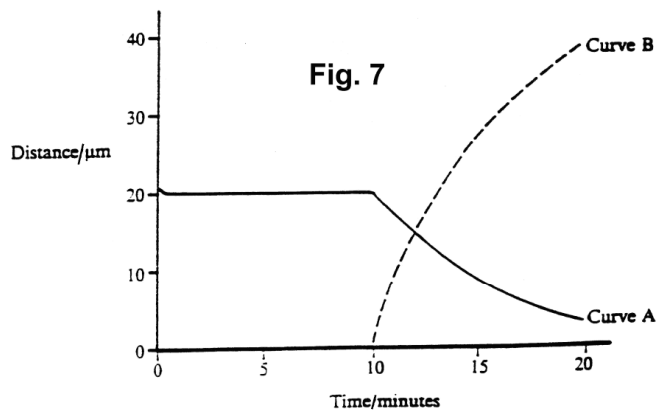
- A. nucleus
- B. mitochondrions
- C. Golgi-complex
- D. endoplasmic reticulum
- E. microtubuli

17.(2pt) Figure 7 shows the movement of chromosomes during mitosis.

Curve A in figure 7 shows the mean distance between the centromeres of the chromosomes and the corresponding pole of the spindle.

At what time did the anaphase start?

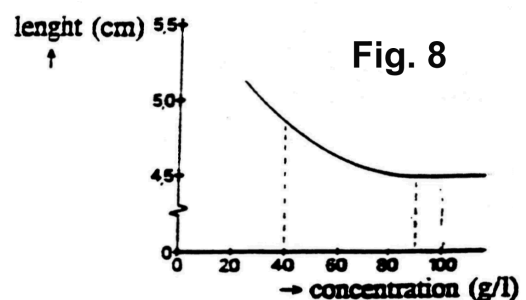
- A. 0 minutes
- B. Somewhere between 0 and 10 minutes
- C. 10 minutes
- D. After more than 10 minutes



- 18.(2pt) In figure 7, curve B represents the distance between
- A. the chromosomes in the metaphase
 - B. the centrioles
 - C. chromatides of a chromosome
 - D. homologous chromosomes during the anaphase
- 19.(1pt) What property/properties make(s) the phospholipids especially well suited to make up cellular membranes?
- A. they are hydrophobic
 - B. they are hydrophilic
 - C. they absorb water readily
 - D. they are both hydrophobic and hydrophilic
- 20.(1pt) If one were observing a cell undergoing mitosis, which of the following would be evidence that it was a plant rather than an animal cell?
- A. absence of spindle
 - B. absence of centrioles
 - C. replication of chromosomes
 - D. disappearance of nuclear membrane
- 21.(1pt) The belonging of a human erythrocyte to serotypes A, B, O is determined by chemical markers on its surface. These markers are:
- A. lipid molecules
 - B. oligosaccharides
 - C. polypeptides
 - D. antibodies
 - E. nucleic acids
- 22.(1pt) Most plants store energy in the form of:
- A. glycogen
 - B. cellulose
 - C. fat
 - D. glucose
 - E. starch
- 23.(1pt) Which organelles do not have a membrane?
- A. mitochondrions
 - B. plastides
 - C. Golgi-apparatus
 - D. nucleolus
 - E. lysosomes
- 24.(1pt) DNA replicates during:
- A. G₁ phase
 - B. S phase
 - C. G₂ phase
 - D. M phase

- 25.(1pt) The molecular formula of glucose is $C_6H_{12}O_6$. What would be the molecular formula of a polymer made by linking ten glucose molecules together by a condensation reaction?
- $C_{60}H_{120}O_{60}$
 - $C_6H_{12}O_6$
 - $C_{60}H_{102}O_{51}$
 - $C_{60}H_{100}O_{50}$
 - $C_{60}H_{112}O_{51}$
- 26.(1pt) Cells of the pancreas will incorporate radioactive amino acids into proteins. This labels newly synthesized proteins and enables to track the location of these proteins in a cell. In this case we are tracking an enzyme that is eventually secreted by pancreas cells. Which of the following is the most likely pathway for movement of this protein in the cell?
- endoplasmic reticulum - Golgi $\rightarrow$ nucleus
 - Golgi - endoplasmic reticulum $\rightarrow$ lysosome
 - nucleus - endoplasmic reticulum $\rightarrow$ Golgi
 - endoplasmic reticulum - Golgi $\rightarrow$ vesicle that fuses with plasma membrane
 - endoplasmic reticulum -lysosome $\rightarrow$ vesicles that fuses with plasma membrane
- 27.(1pt) A chromosome is in its most extended form during:
- interphase
 - prophase
 - metaphase
 - anaphase
 - telophase
- 28.(1pt) Which one of the following elements is needed for the synthesis of chlorophyll, but is not itself a component of chlorophyll?
- carbon
 - magnesium
 - nitrogen
 - iron
 - oxygen
- 29.(1pt) With the help of a light microscope, it can be observed that when cells are dehydrated, plant cells shrink less than animal cells. This is caused by the difference in:
- the flexibility of the plasmalemma
 - the number of pores in the plasmalemma
 - the osmotic potential of the cells
 - the existence of a cell wall in plant cells
 - the size of the vacuoles

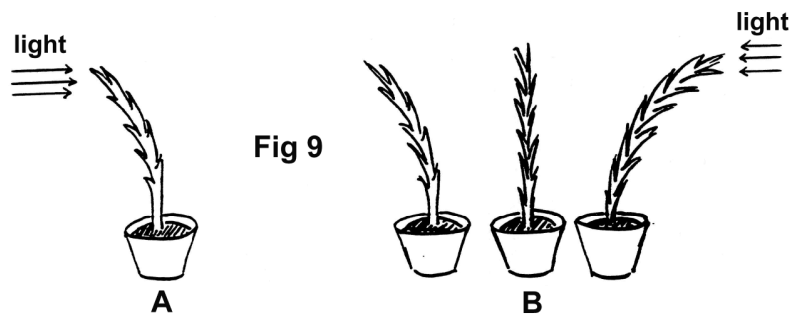
- 30.(2pt) From a fresh potato, small lengths are cut which are exactly 5 cm. Each one of these lengths is placed in a mannitol solution. The concentration of the solution is different for each length. After 4 hours, the length of each potato part is measured. In figure 8, the lengths of the parts are given and compared with the concentrations of the mannitol solutions.



Which of the following conclusion(s) is/are correct?

- I. The cells of the potato length in a solution of 40 grams of mannitol per litre still have turgor after 4 hours.
 - II. The cells of the potato length in a solution of 100 grams of mannitol per litre have come loose from the cell wall after 4 hours.
- A. both
 - B. I only
 - C. II only
 - D. neither

- 31.(2pt) If a stem is exposed to light from one side only, it bends towards the light (Fig.9A). This positive phototropism is caused by substances which are synthesized in the top of the stem. In a stem of a plant (which had been exposed from all sides) a crosswise incision was made just under the top. In the incision, a small slide of glass was placed, after which the plant was exposed on the side of the incision (Fig.9B).



How will the stem now grow, and what is the correct explanation of this phenomenon?

- A. The stem will bend away from the light. The sheet of glass in the incision causes an increase in the concentration of growth-stimulating substances on the exposed side.
- B. The stem will bend away from the light. The sheet of glass stops the transport of growth-inhibiting substances on the exposed side.
- C. The stem keeps growing straight upward. Positive phototropism of the untreated plant is not caused by a decrease in the growth on the exposed side.
- D. The stem will bend towards the light. Light causes the transport of growth stimulating compounds towards the unexposed side of the stem.

- 32.(1pt) Four types of Phytoplankton (I, II, III and IV) were collected from different depths of the sea. For each of these types, the photosynthesis was measured, as represented in figure 10.

Which type of Phytoplankton was collected at the greatest depth?

- A. I
- B. II
- C. III
- D. IV

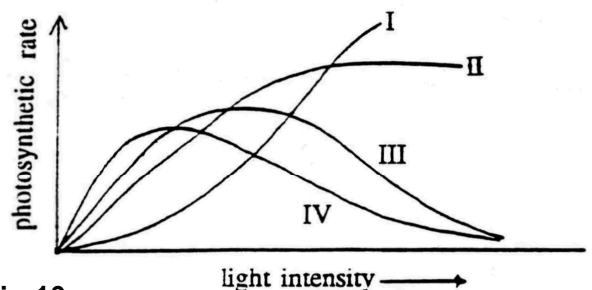


Fig 10

- 33.(1pt) Figure 11 shows leaves of the same type of plant in different conditions. From the edges of the leaves of plant P, small drops of water emerge, from plant Q they do not.

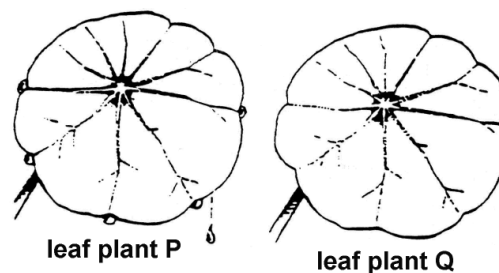


Fig. 11

From which plants do the leaves have the strongest osmotic potential?
From which plant do the leaf cells have the highest turgor pressure?

- | | | |
|----|------------------------------|--------------------------|
| | strongest osmotic potential: | highest turgor pressure: |
| A. | plant P | plant P |
| B. | plant P | plant Q |
| C. | plant Q | plant P |
| D. | plant Q | plant Q |
- 34.(1pt) A C₃- and a C₄-plant are both put into the same glass jar. The jar is kept away from the outside air and is exposed to light during 12 hours a day for some time. After a week the C₃-plant has died. This happens because:
- A. the C₄-plant needs less oxygen for respiration at night
B. the C₄-plant is more resistant to drought
C. the C₄-plant is more efficient in its photosynthesis
D. the C₄-plant exhausts the C₃-plant by photosynthesizing its carbon dioxide which is released during respiration
- 35.(1pt) Figure 12 gives information on the diameter of a tree-trunk during a number of subsequent twenty-four-hour periods.

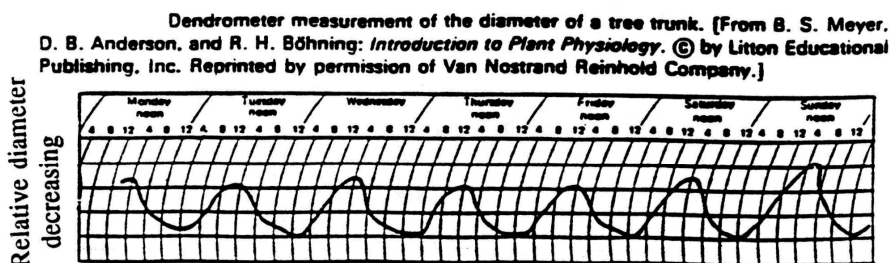


Fig.12

Friday was a dry, cloudy day.

What do you expect the weather conditions were like on the last day?

- A. rainy and calm
B. rainy with a lot of wind
C. sunny day with high humidity
D. sunny day with low humidity
- 36.(1pt) Potassium plays an important role in opening and closing of stomata. This role consists of:
- A. changing the permeability of the plasmalemma of the guard cells for water
B. changing the osmotic potential of the guard cells
C. regulating the synthesis of the plant hormone called abscisic acid, which is needed for the conversion of sugars into starch in the guard cells.

D. regulating the production and degradation of starch and organic acids in the guard cells

- 37.(1pt) Under conditions of a high atmospheric humidity hardly any calcium (Ca) is transported to developing fruits. This is caused by:
- A. calcium only being transported through the xylem and this transport not taking place anymore
 - B. calcium only being transported through the phloem and this transport not taking place anymore
 - C. transpiration stopping and, as a result both xylem and phloem transport stopping
 - D. the stomata closing and transport to the fruit stopping

- 38.(1pt) C₄-plants can start photosynthesis with a lower concentration of CO₂ in the atmosphere than C₃-plants. This is because:
- A. respiration of C₄-plants is higher
 - B. respiration of C₄-plants is lower
 - C. C₄-plants do not have photorespiration
 - D. C₄-plants do have photorespiration

- 39.(1pt) In figure 13, for two plants the relation is shown between the day length and the time at which flowering occurs.

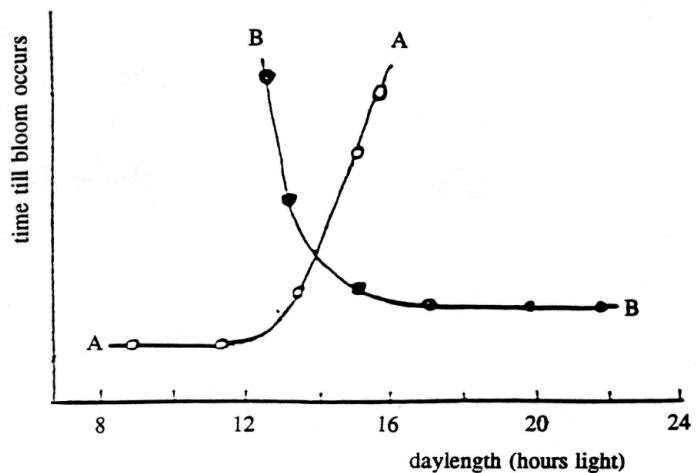


Fig.13

- A. plant A is a short-day plant and plant B is a long-day plant
 - B. plant A is a long-day plant and plant B is a short-day plant
 - C. plant A is a short-day plant and plant B is a day-neutral plant
 - D. plant A is a long-day plant and plant B is a day-neutral plant
 - E. plant A is a day-neutral plant and plant B is a short-day plant
- 40.(1pt) From two trees, diameter 10 and 40 cm, the bark of the stem is equal in thickness. The reason for this is:
- I. secondary phloem is produced much less than secondary xylem
 - II. dead phloem cells are pressed together
 - III. rhytidome is produced and the outer bark peels off
- What is correct ?
- A. I and II only
 - B. I and III only
 - C. II and III only
 - D. I, II and III

- 41.(1pt) The table in Fig. 14 shows the percentages of identity of alpha-hemoglobin amino acid sequences among four animal species: L, M, N, P.

	L	M	N	P
L	100	90	82	97
M	90	100	82	90
N	82	82	100	82
P	97	90	82	100

Fig.14

The species investigated are: man, chimpanzee, horse and kangaroo, but it is not known which letter corresponds to which species. Which one of the given keys is most likely to be correct?

	man	chimpanzee	horse	kangaroo
A.	L	M	N	P
B.	P	N	M	L
C.	L	P	M	N
D.	M	N	L	P

- 42.(1pt) In one of the groups listed below more than one heart may be found. In which group?
- echinoderms
 - bivalves
 - cephalopods
 - non-vertebrate chordates
- 43.(1pt) Only one of the following features of the phylum of the Chordata also is present in adult Tunicata (=Urochordata). Which feature?
- possession of a chorda
 - possession of visceral slits (= pharyngeal slits)
 - possession of a tail
 - possession of a dorsal tubular nervous system
- 44.(1pt) One of the following properties can be used to distinguish reptiles from amphibians. Which one?
- a closed type of circulatory system
 - reproductive organs opening into the intestine
 - a single compartment stomach
 - a metanephros type of kidney
- 45.(1pt) The increase in complexity of the vertebrate circulatory system is represented by one of the following combinations. Which combination?
- toad - rabbit - alligator - shark
 - shark - frog - alligator - rabbit
 - shark - crocodile - rabbit - frog
 - alligator - dog - shark - toad

- 46.(1pt) Blood loses its oxygen rapidly in one of the following places. Which one?
- lung alveoles
 - blood capillaries in intestines
 - lung veins
 - liver portal vein
- 47.(1pt) Four possibilities for the transport of carbon dioxide from the body cells to the lungs are listed below. Which possibility does not exist?
- bound to the ferro-ions of hemoglobin in erythrocytes
 - as a hydrocarbonate ion in the buffering system of the blood
 - bound to the protein of hemoglobin in erythrocytes
 - dissolved in blood plasma and in erythrocyte cytoplasm
- 48.(1pt) Which process in the kidney nephron is least selective?
- secretion
 - reabsorption
 - filtration
 - transport across the epithelium of the collecting duct
- 49.(1pt) The concentration of a substance in Bowman's capsules of a healthy human kidney is 0.1%, while its concentration in the urine is zero. Which one of the following substances is meant?
- calcium phosphate
 - glucose
 - sodium chloride
 - uric acid
- 50.(1pt) Which of the four organs listed below does not belong to the immune system?
- lymph nodes
 - pancreas
 - thymus
 - spleen
- 51.(1pt) Which of the following statements about macrophages is false?
- Macrophages look like amoebas.
 - Macrophages produce antibodies.
 - Macrophages produce interleukines
 - Macrophages practise phagocytosis
- 52.(1pt) Which organ(s) in Fig. 15 is/are under the control of a hormone released by the pituitary gland?
- 1 only
 - 2 only
 - 1 and 2 only
 - 2 and 8 only
 - 1, 2 and 8

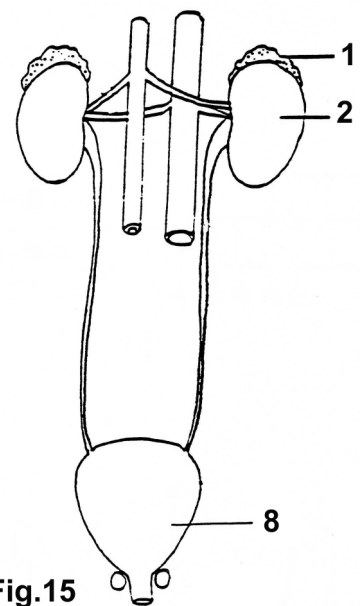


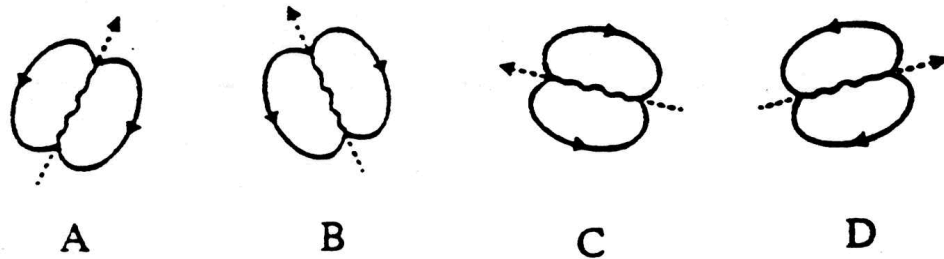
Fig.15

- 53.(1pt) Which one of the following symptoms can be used as evidence for an insufficiency of the hormones secreted by the parathyroid glands?
- decrease in blood glucose
 - increase in blood urea
 - decrease in blood calcium
 - decrease in blood amino acids
- 54.(1pt) The vitelline sac (=yolk sac) is expected to be very small in one of the following groups. In which one?
- in groups that fertilize externally
 - in groups with embryos that are fed from maternal blood
 - in groups that fertilize internally
 - in groups that have an allantoic membrane
- 55.(1pt) One of the following traits is common to both avian and mammalian development. Which one?
- holoblastic cleavage
 - primitive streak
 - trophoblast
 - yolk plug/mass
- 56.(1pt) Ecto-, endo- and mesoderm develop into tissues and organs. One of the following combinations is true. Which one?
- | | ectoderm | mesoderm | entoderm |
|----|-----------------------|--------------------------|----------|
| A. | brain and spinal cord | blood | lungs |
| B. | brain and spinal cord | colon (=large intestine) | lungs |
| C. | skin | bones | kidneys |
| D. | epidermis | liver | heart |
- 57.(1pt) One of the alternatives below defines the layers of the retina in the correct sequence. Which one?
N.B! The first layer in each sequence is supposed to be located next to the jellylike vitreous humor that fills the eyeball.
- pigmented cells - bipolar cells - ganglion cells - photoreceptors
 - photoreceptors - pigmented cells - ganglion cells - bipolar cells
 - ganglion cells - bipolar cells - photoreceptors - pigmented cells
 - photoreceptors - bipolar cells - ganglion cells - pigmented cells
- 58.(1pt) During starvation or hibernation, the supplies of energy substrates are used up in the order:
- fats - proteins - carbohydrates
 - fats - carbohydrates - proteins
 - carbohydrates - fats - proteins
 - proteins - carbohydrates - fats
- 59.(1pt) One of the four statements given below is false. Which one?
- Certain innate patterns of behaviour can be modified by experience.
 - Taxis never occurs together with fixed action patterns.
 - A super normal stimulus often produces a stronger response.
 - Fixed action patterns are highly stereotyped instinctive behaviours.

60.(1pt) The red patch of the bill of a herring gull has a special function for its chickens. One of the statements below is true. Which one? It serves as a:

- A. super normal stimulus
- B. sign stimulus
- C. inhibitor of aggression
- D. recognition marker of the parent

61.(1pt) Some bees in an observation hive found a white clover field at 500 m. distance. At 10 a.m. they danced in the hive according to figure 16A.



How would these bees dance at 2 p.m.?

- A. according to figure 16A
- B. according to figure 16B
- C. according to figure 16C
- D. according to figure 16D

62.(1pt) In the same hive other bees found a group of linden trees; they danced at 10 a.m. according to figure 16B. How would these bees dance at 2 p.m.?

- A. according to figure 16A
- B. according to figure 16B
- C. according to figure 16C
- D. according to figure 16D

63.(1pt) A number of moth species use pheromones to achieve the encounter between the sexes. Generally it is the stationary female that produces this odour, and the male often flies long distances to find the female.

- A. He is activated by the pheromones and flies upwind to the odour source (anemotaxis); the pheromones induce landing near the female.
- B. He is activated by the pheromones and uses the odour gradient (chemotaxis) both for orientation and landing.
- C. He is activated by climatic conditions; after a random flight he spots the female nearby through the perception of pheromones.
- D. He is activated by climatic conditions; celestial cues enable a directed flight; the pheromones induce landing near the female.

64.(1pt) During autumn migration the young of some bird species fly with their parents, while in other species old and young birds migrate separately. This is related to :

- A. the number of broods the parents make in a season
- B. the average life span of an individual of that species
- C. food requirements during migration
- D. the direction and distance of the migration being learned or innate

- 65.(1pt) Pigeons were trained to collect food from a feeding machine at 10 a.m., by picking at a pigeon key. Then they were transported 60 degrees eastward to a place P. At this new location they are expected to pick at the pigeon key (local time in P) at:
- A. 10 a.m.
 - B. 4 a.m.
 - C. 4 p.m.
 - D. 2 p.m.
- 66.(1pt) The most important factor regulating seasonal migration is:
- A. the change in average air temperature
 - B. the change in day length
 - C. the reduced availability of food
 - D. the increased predator pressure
- 67.(1pt) Which of the following alternatives is not an indication of annual photoperiodism?
- A. swarming
 - B. moulting
 - C. flowering
 - D. sleep
- 68.(1pt) So-called altruistic behaviour (e.g. the protection of lion pups by 'aunts') occurs in natural populations because (choose the primary reason):
- A. it ensures survival of the species
 - B. it ensures survival of kin
 - C. it ensures spread of the genes shared by relatives
 - D. of group selection mechanisms
- 69.(1pt) As basic material for micro evolution serve(s)
- A. modifications
 - B. phenotypical plasticity
 - C. mutations
 - D. phenocopies
- 70.(2pt) In *Drosophila*, the following crossing-over percentages were found:
- | gene | gene | crossing-over % |
|------|------|-----------------|
| bi | ec | 1.4 |
| bi | fa | 3.9 |
| wi | ec | 4.0 |
| wi | fa | 1.5 |
- What is the order of these genes?
- A. bi - ec - fa - wi
 - B. bi - ec - wi - fa
 - C. ec - bi - fa - wi
 - D. ec - bi - wi - fa
- 71.(1pt) Which form of natural selection can lead to differentiation in species and to polymorphism?
- A. directional selection
 - B. stabilizing selection
 - C. disruptive selection
 - D. density dependent selection

- 72.(1pt) What is the advantage of sexual reproduction over a-sexual reproduction?
Sexual reproduction generally leads to:
- A. a larger amount of offspring
 - B. greater genetic variation among the offspring
 - C. a greater distribution area
 - D. lower sensitivity of the individual to environmental influences
- 73.(1pt) A black female rabbit which had been crossbred with a white male rabbit had a litter of black young. One black young female from this litter was crossbred with a white male. What is to be expected of the colours of young rabbits from this second crossbreeding when, in this case, fur colour is determined by an autosomal locus?
- A. 3 blacks and 1 white
 - B. 3 whites and 1 black
 - C. blacks only
 - D. 2 whites and 2 blacks
- 74.(1pt) Which new qualities in the organic world occurred in the Precambrium:
- A. the heart with four ventricles and homeothermia
 - B. the flower and seeds
 - C. photosynthesis and multicellularity
 - D. internal skeleton of bone tissue
- 75.(1pt) Which changes in the organic world occurred in the Mesozoic:
- A. development and spreading of the angiosperms
 - B. development and spreading of the reptiles
 - C. development and spreading of fish
 - D. development and spreading of the green algae
- 76.(1pt) Which factor possibly played a selecting role in the evolution of the flower:
- A. the fertilization process becoming independent on the presence of water
 - B. development of adaptations to pollination
 - C. development of a surplus of nutrients in the seed
 - D. development of adaptations to the spreading of fruits
- 77.(1pt) What did the evolution of photosynthesis lead to?
- A. the evolution of the biosynthesis of proteins
 - B. the evolution of multicellular organisms
 - C. the accumulation of carbon dioxide in the atmosphere
 - D. the accumulation of oxygen in the atmosphere
- 78.(1pt) In which of the following cases the genes perform a multiple function?
- A. A pea has a dominant gene which causes the yellow colour of the seeds.
 - B. The green colour of the seeds of a pea is caused by a recessive gene.
 - C. Yellow body colour of fruit flies is inherited sex-linked.
 - D. With the columbine (flower genus *Aquilegia*), a gene determines the red colour of the flower and the violet hue in the leaves.

79.(1pt) How are polyploid organisms obtained:

- A. Self-pollination is carried out with plants and homozygous lines are obtained; the homozygous lines are crossbred among themselves and show heterosis.
- B. Individuals are selected which are of interest to man, their sex organs are exposed to radiation and as a result of changes in genes and chromosomes new hereditary characteristics occur.
- C. For a good productivity, a selection is made in a number of lines, and these are then crossed.
- D. By means of a colchicine treatment, the spindle of a dividing cell collapses and therefore, the doubled chromosomes do not disperse towards the poles of the meiotic cell.

80.(1pt) Which two pictures in figure 17 give a summary of the theory of natural selection according to Darwin:

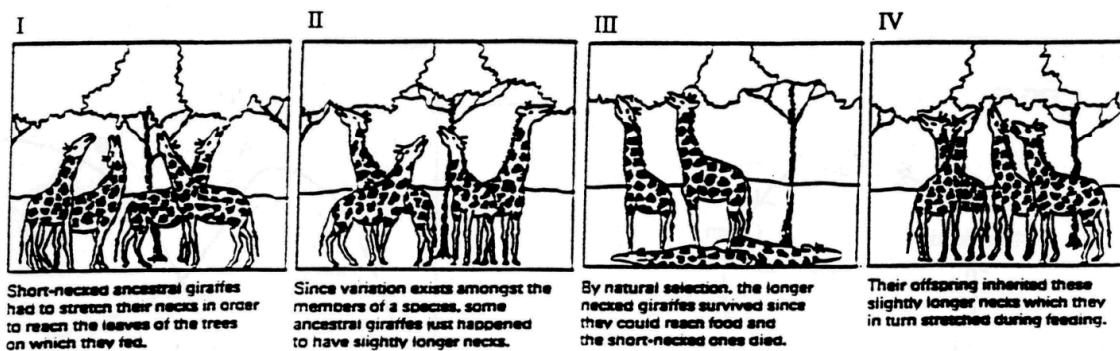


Fig. 17 Some pictures of possible intermediate stages in the evolution of the modern giraffe.

- A. I and III
- B. I and IV
- C. II and III
- D. II and IV

81.(1pt) In the European population, about 1 in 2500 people suffers from Cystic Fibrosis, a genetically determined (descared), autosomal disease. Healthy parents have a child suffering from Cystic Fibrosis. The woman remarries a healthy man. What is the chance of a child from this second marriage suffering from Cystic Fibrosis:

- A. 1 : 25
- B. 1 : 50
- C. 1 : 100
- D. 1 : 625

82.(2pt) The pedigree in figure 18 shows an inheritance of a rare form of muscular dystrophy.

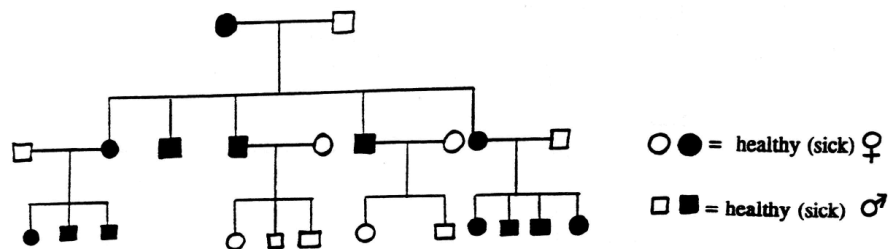


Fig. 18

The disease is probably caused by a mutation on one locus which is:

- A. recessive, autosomal
- B. dominant, autosomal
- C. recessive, related to the X-chromosome
- D. related to the Y-chromosome
- E. situated in the mitochondrial genome

- 83.(1pt) Extinct marine xiphosaurs are found in a large area in 150 million-year-old sediments (150 Myr BP). From each individual, the length/breadth ratio of the carapax was measured. This length/breadth ratio is referred to as 's'.
In figure 19, graph P represents the distribution of quality 's' for 150 Myr BP.

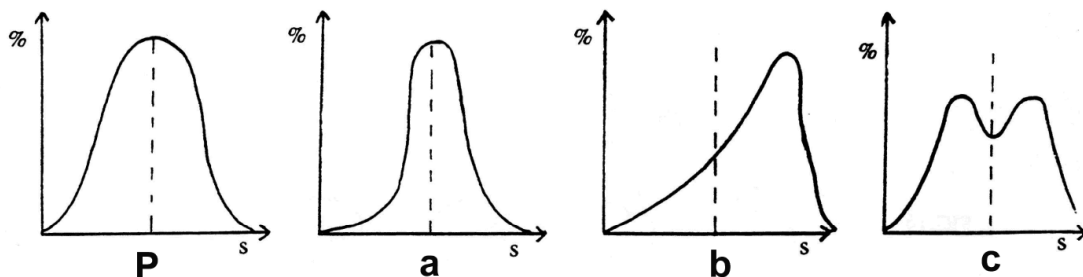


Fig. 19

In sediments of 100 Myr BP, three different populations of xiphosaurs are found in three different places. The graphs a, b, c in figure 19 represent the distribution of quality 's' in each of these three populations. Answer the following two questions:

- 1 for which of the three populations (a, b, c) is it most probable that species formation occurs?
- 2 for which of the three areas (a, b, c) is it most probable that the environment remained constant?

	species formation	constant environment
A	b	a
B	c	c
C	c	a
D	b	b

- 84.(1pt) The order of appearance of the main groups of organisms during evolution can best be described as follows:
- A. autotrophic, anaerobic heterotrophic, aerobic heterotrophic
 - B. aerobic heterotrophic, anaerobic heterotrophic, autotrophic
 - C. anaerobic heterotrophic, aerobic heterotrophic, water oxydizing autotrophic
 - D. anaerobic heterotrophic, water oxydizing autotrophic, aerobic heterotrophic
- 85.(1pt) The main trend in the evolution of land plant was:
- A. a sharp demarcation of the phases of sporophyte and gametophyte
 - B. a shortening of the haploid phase
 - C. a shortening of the a-sexual phase
 - D. an increase in the complexity of the gametophyte

- 86.(1pt) A new characteristic usually appears in evolution as a result of:
- accumulation of point mutations in a gene which originally encoded for something else
 - duplication of a gene and accumulation of point mutations in one of the copies coming from that duplication
 - a mutation in a regulator gene
 - genotypical recording of favourable phenotypical adaptations

- 87.(2pt) How many different phenotypes can be expected in the F_2 of the crossing: AA BB * aa bb when:
I the genes are completely coupled and,
II the genes inherit independently:

	I	II
A.	3	4
B.	3	9
C.	4	9
D.	4	16
E.	9	16

- 88.(3pt) In the domestic cat, the locus Orange is sex linked and intermediary dominant: heterozygous cats are 'tortoise'. A tortoise pussy cat (Oo) mates with a red tomcat (oY); the third colour is tabby (OO, OY). What can be expected for the fur colour of the kittens?

	females	males
A.	tortoise	red and tabby
B.	tortoise and red	red and tortoise
C.	tortoise and red	red and tabby
D.	red and tabby	red and tabby

- 89.(3pt) In the domestic cat, the autosomal locus White is dominant and epistatic; the locus Orange is sex linked with allele O tabby and allele o red, while the heterozygous is tortoise. A white female mates with a tabby tomcat.

The kittens turn out to be:

- 1 red male
- 1 tortoise female
- 1 tabby female
- 1 white male
- 1 white female

What is the genotype of the mother:

- WW Oo
- Ww OO
- Ww Oo
- Ww oo

- 90.(1pt) The following reactions produce the eye pigments B and D in fruit flies:

enzyme X

substrate A -----> pigment B

enzyme Y

substrate C -----> pigment D

Mixing of both pigments results in a reddish brown eye colour with wild type flies. Absence of pigment B results in bright red eyes, absence of pigment D results in brown eyes, absence of both pigments results in colourless (white) eyes.

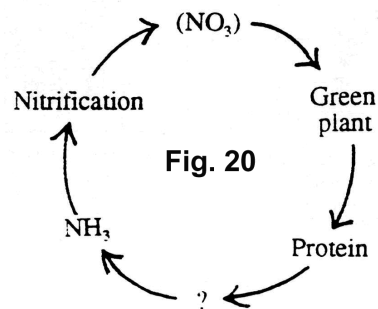
When a homozygous fly with colourless eyes is crossed with a homozygous wild type fly, which enzymes can be demonstrated in the F_1 descendants:

- A. X only
- B. Y only
- C. X and Y
- D. none

- 91.(3pt) Crossing of two white dogs with the same genotype results in descendants in the following relation: 12 white : 3 black : 1 brown. When the brown descendant is crossed with a white which has the same genotype as the parents, what is the expected relation in this second crossing:

	white :	black :	dark brown :	brown
A.	9	3	3	1
B.	1	1	1	1
C.	12	3	-	1
D.	2	1	-	1

- 92.(1pt) A simplified representation of the nitrogen cycle in an ecosystem is shown in figure 20. Which group of bacteria is involved in the conversion of proteins to NH_3 ?



- A. Decomposers
 - B. Chemosynthetizers
 - C. Nitrogen oxidizers
 - D. Free nitrogen fixers
 - E. Ammonia oxidizers
- 93.(1pt) Nitrogen is fixed in ecosystems in ways stated below. One of the statements below is false. Which one?
- A. by cyanobacteria
 - B. by electrical discharges in the atmosphere
 - C. by industrially synthesized fertilizer
 - D. by atmospheric deposition
 - E. by denitrification
- 94.(1pt) Water in an eutrophic, fresh-water lake, contains large amounts of blue-green algae, which make the lake unsuitable for recreation. Which of the following treatments is the most likely to lead, within the shortest time, to a significant decrease in the abundance of these algae ?
- A. Massive introduction of carnivorous fish (bass, pike etc.).
 - B. Massive introduction of zooplankton eating fish.
 - C. Complete removal (by selective fishing) of carnivorous fish.
 - D. Complete removal (by selective fishing) of zooplankton eating fish.

- 95.(1pt) The transfer of energy through a terrestrial ecosystem is often depicted by energy pyramids. Which of the following statements is true ?
- A. Ecological efficiency is highest for top consumers.
 - B. About 10 % of the energy from one trophic level is incorporated into biomass of the next level.
 - C. The energy lost as heat or in cellular respiration is 10 % of the available energy of each trophic level.
 - D. Only 25 % of the energy in one trophic level is passed on to the next level.

- 96.(1pt) Nitrogen fertilization of rice fields can be low in the presence of one of the following types of organisms. Which one ?
- A. Water fern (*Azolla* species)
 - B. Green algae
 - C. Brown algae
 - D. Mosses

- 97.(1pt) On the border of two ecosystems (grassland and wood), there is a transitional zone (ecotope). Characteristic for the fauna of this ecotope is:
- A. A smaller species diversity in comparison with the two adjoining ecosystems.
 - B. A greater species diversity in comparison with the two adjoining ecosystems.
 - C. A greater number of herbivores in comparison with the two adjoining ecosystems.
 - D. A greater number of predators and parasites in comparison with the two adjoining ecosystems.

- 98.(1pt) The growth of a population of animals is mainly determined by the combination of two important parameters. Which?
- A. birth rate and food supply
 - B. mortality rate and migration
 - C. mortality rate and the size of the area occupied by the population
 - D. birth and mortality rate
 - E. food sources and the size of the territory

- 99.(1pt) Which one of the sequences below is an example of primary succession?
- | | | |
|----------------------|---------------------|---------------------|
| A. mosses | - lichens | - herbaceous plants |
| B. lichens | - herbaceous plants | - mosses |
| C. lichens | - mosses | - herbaceous plants |
| D. herbaceous plants | - mosses | - lichens |
| E. herbaceous plants | - lichens | - mosses |

- 100.(2pt) In the graph of figure 21, for an area in the temperate regions, two curves are shown, which are related to the changes of two quantities during primary succession. What do these curves represent?

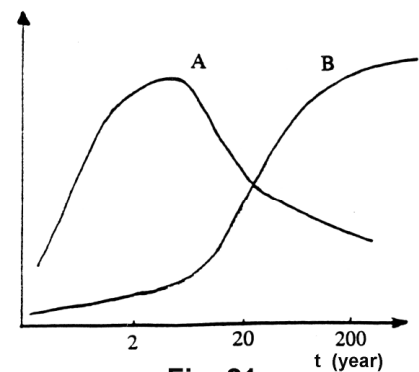


Fig. 21

- A. A = the species diversity of herbaceous plants
B = the biomass of the entire vegetation
- B. A = the biomass of the entire population
B = the species diversity of herbaceous plants
- C. A = the total amount of humus on the soil
B = the species diversity of the entire vegetation
- D. A = the species diversity of the carnivores
B = the total biomass of the carnivores

101.(1pt) In figure 22, the total amount of humus is plotted as a function of time, in two ecosystems.

What are these systems?

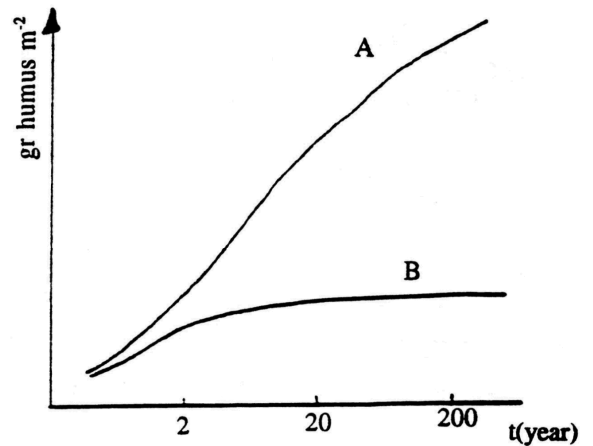


Fig. 22

- | | A | B |
|----|----------------------|--|
| A. | tropical grassland | tropical rain forest |
| B. | tropical rain forest | broad-leaved forest from temperate regions |
| C. | tundra | tropical rain forest |
| D. | tropical grassland | grassland from temperate regions |

102.(1pt) In a laboratory, a competition experiment is executed with two types of bacteria, which compete for food. During the experiment, it is determined what the relationship is between the frequency of type 1 in the mixed culture after 1 generation (z_{t+1}) as a function of the frequency of type 1 during the previous generation (z_t). In figure 23, the drawn line indicates the observed relationship between z_{t+1} and z_t , the dotted line is the relationship $z_{t+1} = z_t$.

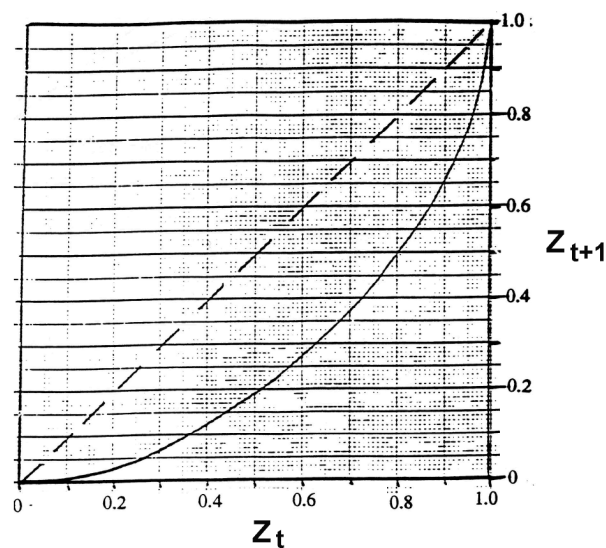


Fig. 23

What happens to type 1 and type 2 in the long term?

- A. Type 1 co-exists with type 2.
- B. Both type 1 and type 2 grow in number.
- C. Type 1 excludes type 2 from the mixed culture.
- D. Type 2 excludes type 1 from the mixed culture.

- 103.(2pt) In a trout hatchery, a density experiment was executed. In figure 24, the eventual density is plotted as a function of the starting density. The eventual densities were determined before a new generation of trouts was born in the fish-breeding pond. What is the explanation for the fact that, for higher starting densities, the eventual density is no longer dependent on the starting densities?

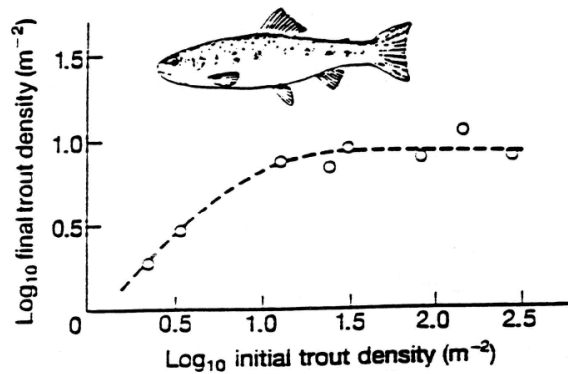


Fig. 24

- A. Accumulation of damaging waste products in the breeding pond.
- B. Exactly compensating density dependent mortality.
- C. Exactly compensating density dependent birth.
- D. Overcompensating density dependent mortality.
- E. Undercompensating density dependent birth.

- 104.(2pt) In figure 25, the biomass of the macrofauna and the microfauna in various ecosystems is presented.

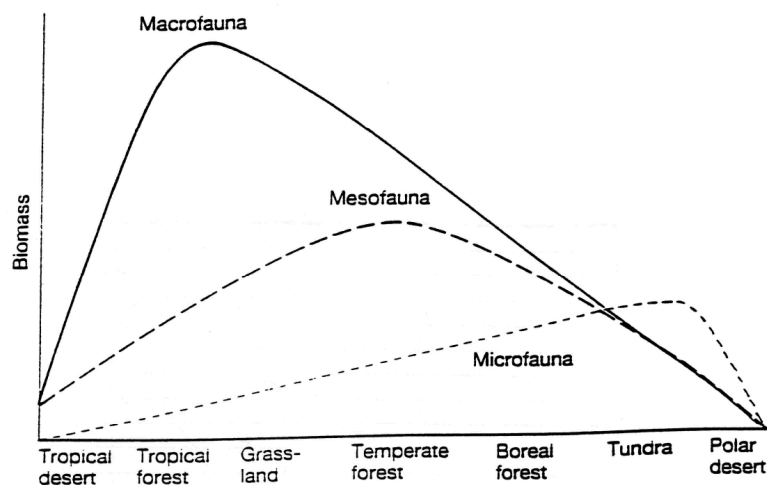


Fig. 25

Which conclusion can be drawn about the breakdown speed of litter and the accumulation rate of humus.

		breakdown rate of litter	accumulation rate of humus
A.	tundra:	low	low
B.	tundra:	low	high
C.	tropical rain forest:	high	high
D.	tropical rain forest:	low	low

105.(2pt) In figure 26, the changes in time of three quantities are shown during the secondary succession after a forest fire.

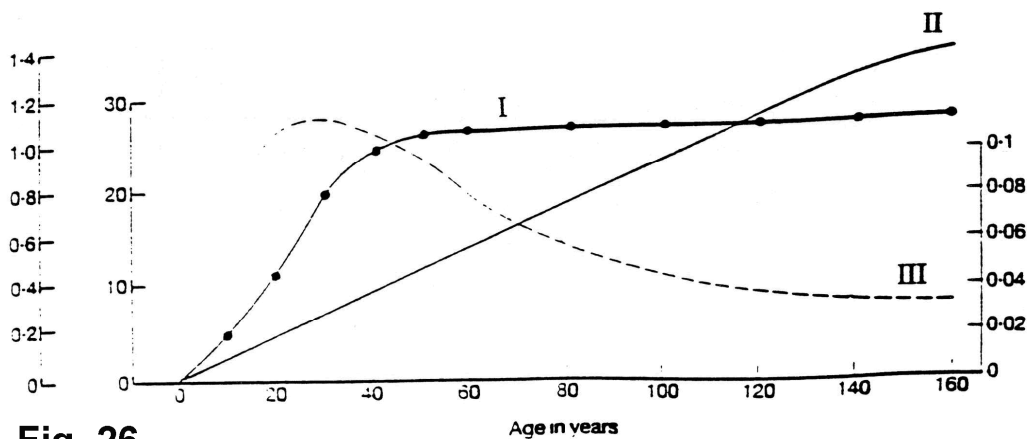


Fig. 26

	A	B	C
A	biomass of the vegetation	net primary production	secondary production
B	net primary production	biomass of the vegetation	ratio between net primary production and the biomass of the vegetation
C	net secondary production	net primary production	net tertiary production
D	ecosystem respiration	net primary production	Biomass of tertiary producers

106.(1pt) During the last decades, the concentration of CO₂ in the atmosphere has gradually increased. Which of the following factors does not partly cause this:

- A. cutting down the tropical rain forest.
- B. The combustion of fossil fuels.
- C. The increasing motorized traffic.
- D. The increasing desert forming in the Sahel zone in Africa.
- E. The world-wide increase of the net primary production.

107.(1pt) Of the following statements, one is not correct. Which one?

- A. Interspecific competition between two species always leads to competitive exclusion of one of the competing species.
- B. Niche differentiation occurs with plants as well as with animals.
- C. Niche differentiation increases the species diversity in an ecosystem.
- D. Self-thinning in plants is the result of intraspecific competition.

- 108.(2pt) The decomposition rate of dead material, expressed as the change in dry weight in time, is often described by means of a negative exponential equation:

$$m_t = m_0 \cdot e^{-kt}$$

In this equation, m_t is the mass at time t , m_0 is the mass at time 0 (when the material has just died off) and k is the decomposition constant. k varies strongly with different types of material.

In which sequences below the value of k is increasing from left to right ?

- | | | | |
|----|------------------|----------------------------------|---------------------------|
| A. | Wood | - leaves of Prunus (cherry-tree) | - leaves of Fagus (beech) |
| B. | Wood | - leaves of Fagus | - leaves of Prunus |
| C. | Leaves of Prunus | - leaves of Fagus | - wood |
| D. | Leaves of Fagus | - leaves of Prunus | - wood |
| E. | Leaves of Fagus | - wood | - leaves of Prunus |

- 109.(3pt) The growth speed of a population can often be described with the logistic growth equation:

$$dN/dt = rN(K-N)/K$$

In this equation, N is the number of individuals, r is the intrinsic relative rate, and K is the carrying capacity.

According to this equation, the equilibrium number of individuals in the population is determined by:

- r only
- K only
- r and K
- N and K
- r and N

- 110.(2pt) In many experiments it has been observed that the light extinction in a vegetation decreases with an increasing Leaf Area Index (LAI : m^2 leaf per m^2 area), in accordance with this equation:

$$I_d = I_0 \times e^{(-k \times LAI_d)}$$

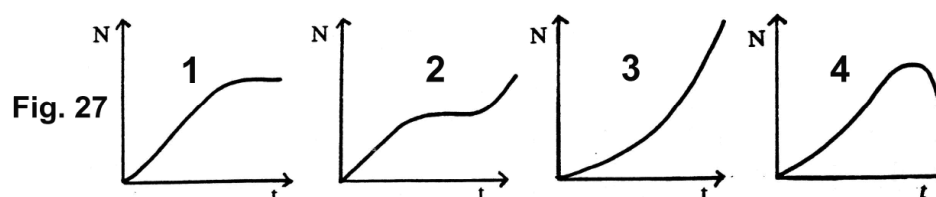
In this equation, I_d is the light intensity at depth d in the vegetation (measured from the top!), I_0 is the light intensity above the vegetation, k is the extinction coefficient and LAI_d the cumulative Leaf Area Index (measured from the top of the vegetation).

Which statement is not correct ?

k increases with an increasing:

- thickness of the leaves
- concentration of chlorophyll in the leaves
- amount of leaves in the vegetation
- reflection of light in the vegetation

- 111.(2pt) The curves 1 to 4 in figure 27 are growth curves of bacterial populations under conditions W, X, Y, and Z.



These conditions are:

- W - continuous supply of food, water, oxygen, removal of waste products
- X - with time, food is depleted and harmful waste products accumulate
- Y - with time, food is depleted and harmless waste products accumulate
- Z - with time, food is depleted; as a result of a mutation, a new strain develops that is capable of metabolizing the waste products secreted by the original strain

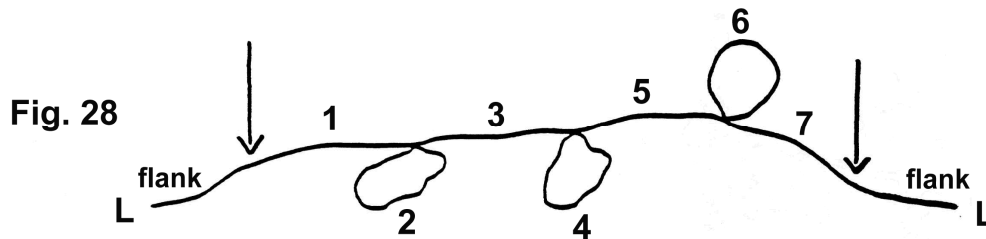
Which growth curves and growth conditions do match correctly ?

	1	2	3	4
A.	W	X	Y	Z
B.	X	Y	Z	W
C.	Y	Z	W	X
D.	Y	X	Z	W
E.	X	Y	W	Z

End of Part A

Theoretical test (Part B)

- 112.(3pt) Chromosomal DNA, on which gene A is located, is isolated. One of the two complementary DNA chains is, after separation, paired with gene A mRNA. This happens under circumstances favouring DNA/RNA hybridisation. After hybridisation a structure as shown in figure 28 can be observed.



The arrows mark the first and last codon for gene A. Just a small portion of the flanking DNA is shown. Indicate in the table which chainpart and which number do match correctly. (ss = single stranded, ds = double stranded).

chainpart:	number:
ss DNA	
ss RNA	
ds DNA	
ds RNA	
ds DNA/RNA	

- 113.(4pt) In the model of a gene given below, the organisation of the DNA in and around the chromosomal "single copy" gene for the polypeptide musculase is shown. Marked are the initiation and termination sites of transcription, the start and stop codons of translation and the boundaries of the intron in the gene (↓). The distances are given in kilobases (kb), but not drawn on scale.

ini	sta	↓	↓	sto	ter	
0	1,2	1,7	2,0	5,2	5,8	7,5 8,0

- A Of how many aminoacids does the polypeptide musculase consist? Assume the musculase does not undergo any posttranslational processing.
- Answer:
- B Of how many nucleotides does the musculase mRNA consist that is used in the ribosomes? Assume that there is a 100 nucleotides long poly-A-tail connected to the mRNA at the 3'-terminus before it is operational.

Answer:

- 114.(4pt) In the wild type of *Escherichia coli* the polypeptide P consists of 169 aminoacids. The sequence of the aa 161 to 165 (counted from the N-terminus) is:

161 162 163 164 165
 --- trp his met glu tyr ---

There is a population which has a mutation in the structural gene for polypeptide P, as a result of which the pp only contains 165 amino acids. The order of the first 161 amino acids is the same as with the wild type. The order of the others is as follows:

161 162 163 164 165
 --- trp thr tyr gly val ---

Assume in the following questions that the mutant mRNA differs in only one nucleotide of the wildtype mRNA.

- a. Give the mRNA codons for the aminoacids 161 to 165 for the wildtype and the mutant. Use figure 1 for the codons. For each aminoacid only one code will be the correct one.

amino acid number	161	162	163	164	165
wild type code	UGG				
mutant code	UGG				

- B. What kind of mutation has happened? Place a tick in the correct column:

substitution mutation	
addition mutation	
deletion mutation	

- C. What is the first letter of the aminoacid 166 in the mutant?

Answer:

- 115.(3pt) Plants can move. We distinguish between nastic movements and tropism. Give the most common mechanism by which these movements take place, both for nastic movements and tropism. Choose from the following list:

- A. by means of contractile proteins
- B. by means of elongation of the group of cells involved
- C. by means of restraining cell division of the group of cells involved
- D. by means of stimulating the cell division of the group of cells involved
- E. by means of changing the permeability of the plasmalemma
- F. by means of water absorption of the group of cell involved

Answer: Tropism:

Nastic movements :

- 116.(3pt) From a one-year-old plant, a growth analysis has been made by measuring a number of parameters at various points in time. In figure 29, the obtained results are plotted on the Y-axis, on different scales, against time (X-axis).

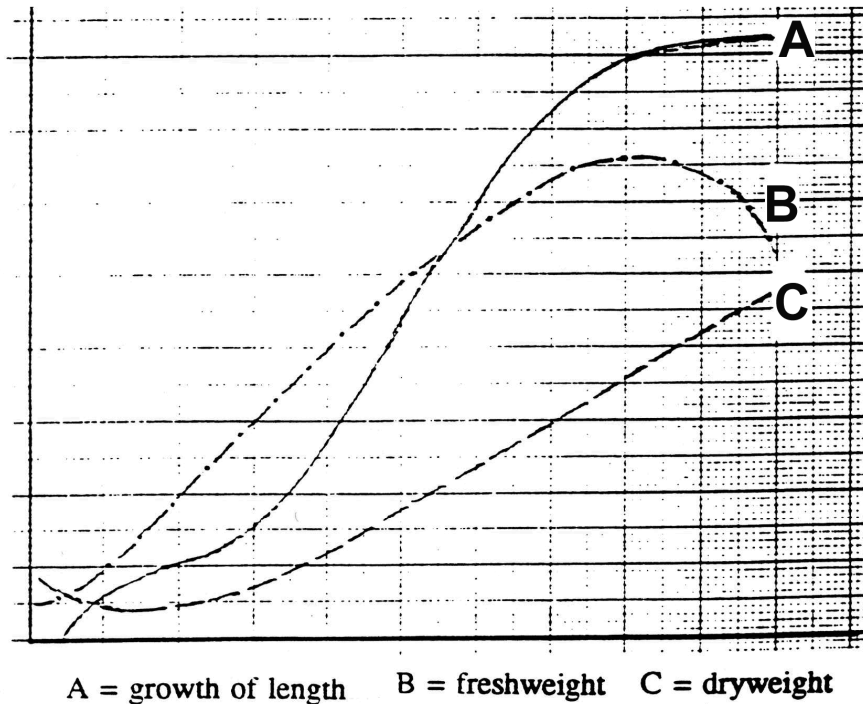


Fig. 29

Match each of the parameters with the curves. Place a tick in the correct column.

	A	B	C
the length growth of the stem	☐	☐	☐
the fresh weight of the whole plant	☐	☐	☐
the dry weight of the whole plant	☐	☐	☐

117.(3pt) Which of the tissues parenchyma, collenchyma and sclerenchyma can have (one of) the characteristics or functions below? Put a tick in the right column.

Fout! Bladwijzer niet gedefinieerd.	parenchyma	collenchyma	sclerenchyma
secondary lignified cell walls	☐	☐	☐
contain chloroplasts	☐	☐	☐
lacking of protoplasts in matured cells	☐	☐	☐
can become meristematic	☐	☐	☐
cells are always elongated	☐	☐	☐

118.(3pt) In the cases indicated in the table, the water absorption of a plant with maximal transpiration rate can change. As a result, the plant will absorb less or more water from the soil, or the water absorption remains the same. Place a tick in the correct column.

Fout! Bladwijzer niet gedefinieerd.	water absorption		
	less	same	more
the soil temperature drops from 20 °C to 5 °C			
the soil dehydrates			
it starts raining			
it gets dark			
25% of the leaves are cut off			
25% of the roots are cut off			

- 119.(2pt) Figure 30 shows the pith rays close to the cambium on a tangential cross-section of the bark of a tree. With this tree, dilatation only takes place in the pith rays.

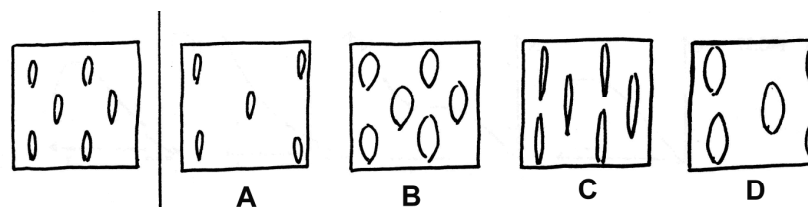


Fig. 30

Quite far from the cambium, the pith rays in the bark, on a tangential cross-section look like figure: (Fill in A, B, C or D)

- 120.(2pt) The soil of an oat field gets exhausted at the beginning of flowering. Then half of the field is manured with artificial fertilizer, which contains all elements needed, and in a form that can easily be taken in by the plants. What will happen with the plants on that part of the field that was not manured ? Place a tick in the correct column.

Fout! Bladwijzer niet gedefinieerd.	true	false
the leaves sooner turn yellow		
the fruits are of less weight		
the fruits have lower protein concentration		
the fruits have a much lower calcium concentration		

- 121.(2pt) The following animal phyla are placed in the supposed chronological order in which they developed during evolution:

- 1-Protozoa;
- 2-Porifera;
- 3-Cnidaria;
- 4-Platyhelminthes;
- 5-Annelida;
- 6-Arthropoda;
- 7-Chordata.

Fill in which of the phyla mentioned above was the first in evolution in having:

- | | | | |
|----|---|-----|-------|
| A. | extracellular digestion? | nr: | |
| B. | a nervous system? | nr: | |
| C. | a closed tubular circulatory system? | nr: | |
| D. | a digestive tract with a mouth and an anus separated from each other? | nr: | |

122.(4pt) Fill in which of the phyla from question 121

- | | | | |
|----|---|------|-------|
| A. | have a ladderlike nervous system? | nrs: | |
| B. | belong to the group of the Coelomata? | nrs: | |
| C. | belong to the group of the Protostomia? | nrs: | |
| D. | belong to the group of the Bilateria (=Bilateralialia)? | nrs: | |

123.(3pt) In the digestive system of the human body, the ingested organic substances are digested (= broken down) to monomers (= simple substances). Indicate the shortest way along which one of these monomers, glucose, can reach the kidneys from the intestine, by putting the numbers of the structures involved in the right sequence.

These numbers are to be chosen from the following list:

- | | | |
|------------------------|----------------------|--------------------------|
| 1 = jugular veins | 7 = aorta | 12 = left ventricle |
| 2 = inferior vena cava | 8 = renal arteries | 13 = carotic arteries |
| 3 = liver | 9 = lungs | 14 = brain |
| 4 = left atrium | 10 = right ventricle | 15 = hepatic portal vein |
| 5 = superior vena cava | 11 = pulmonary veins | 16 = pulmonary arteries. |
| 6 = right atrium | | |

Answer:

124.(2pt) Indicate in the margin with + or — which of the following statements about nutrition and digestion are true (+) or false (—).

- A. In mammals, the protein-hydrolyzing enzyme pepsine is produced in the liver.
- B. In mammals, the production of saliva by the salivary glands is regulated by nerves originating from the brain.
- C. Gland cells in the wall of the stomach of ruminants produce an enzyme that digests cellulose.
- D. Which of the amino acids are to be called exogenous/essential amino acids, as opposed to endogenous/non-essential amino acids, depends on the animal species under consideration.

125.(2pt) Write down the numbers of the following structures as appearing from Fig.31:

See next page

Fout! Bladwijzer niet gedefinieerd.	Answer:
A. distal convoluted tubule:	
B. glomerulus:	
C. collecting tubule (of Bellini):	
D. loop of Henle:	

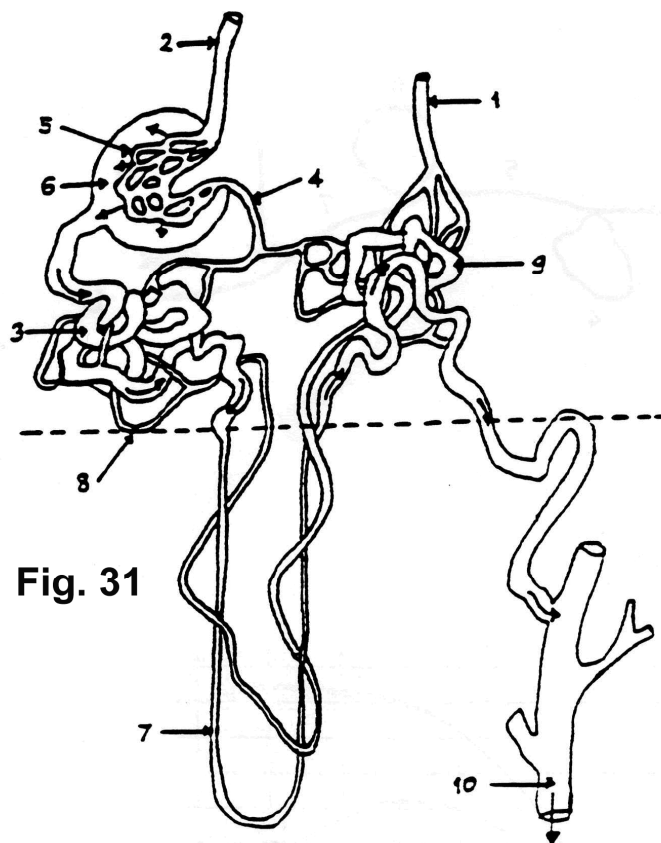


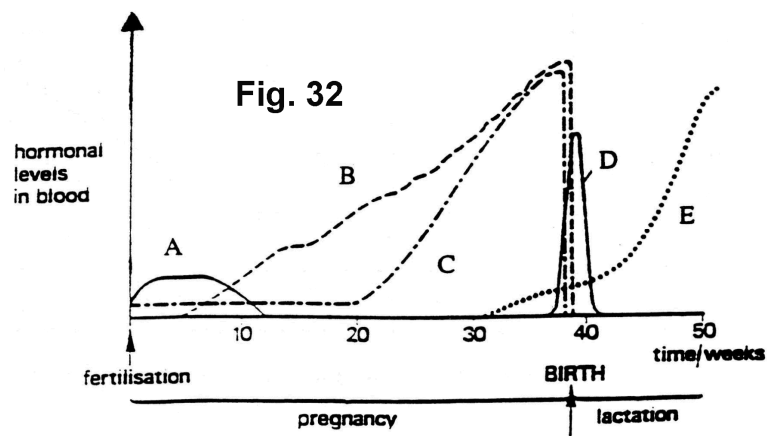
Fig. 31

- 126.(2pt) Rh⁺ blood is transfused into a Rh⁻ woman who has not previously been transfused. Indicate in the margin with + or — which of the following statements are true (+) or false (—).
- A. Anti-Rh antibodies will be produced by the woman.
 - B. The transfused blood and the blood of the woman are incompatible, so red cell agglutination and death may follow.
 - C. In a subsequent pregnancy, a Rh⁺ foetus could be threatened by haemolytic disease.
 - D. There is no immediate or long term effect as 70% of the Rh⁺ population are heterozygous.
- 127.(2pt) Indicate in the margin with + or — which of the following statements about insulin are true (+) or false (—).
- A. Insulin stimulates the conversion of lipids to glucose.
 - B. Excess amounts of insulin in the body can cause a state of coma.
 - C. Insulin increases the storage of glycogen in the liver.
 - D. Insulin increases the blood glucose level.

128.(2pt) Indicate in the margin with + or — which of the following statements about the female reproductive system are true (+) or false (—).

- A. Both oestrogen and progesterone are necessary for ovulation to take place.
- B. Oestrogen inhibits the production of FSH by the anterior pituitary gland.
- C. Fertilization of the ovum by the spermatozoon normally takes place in the uterus.
- D. Progesterone production is largely under the control of LH.

129.(3pt) Figure 32 shows changes in the blood concentration levels of several hormones involved in pregnancy, birth and lactation. The various curves in the Figure are labelled with the letters A to E.



The various hormones involved are indicated in the table below. Which curve belongs to which hormone? Fill in A, B, C, D or E.

Fout! Bladwijzer niet gedefinieerd.hormone:	curve
oestrogen	
oxytocin	
prolactin	
progesterone from placenta	
progesterone from corpus luteum	

130.(2pt) The following statements are related to the regulation of the processes of moulting and metamorphosis during the lifecycle of butterflies.

Indicate in the margin with + or — which statements are true (+) or false (—).

- A. If, early in the second larval stage, the secretion by the corpora allata of juvenile hormone is inhibited; the next moult will result in a new caterpillar.
- B. If, early in the last larval stage, the secretion by the prothoracic glands of ecdyson is inhibited; the metamorphosis to a fully developed butterfly will occur.
- C. If, early in the second larval stage, the secretion by the brain of a hormone which stimulates the prothoric glands is inhibited; there will be no more moulting
- D. If, early in the last larval stage, extra juvenile hormone is injected, the next

moult will result in an extra large caterpillar.

- 131.(2pt) Write down the numbers of the following structures as appearing from figure 33.

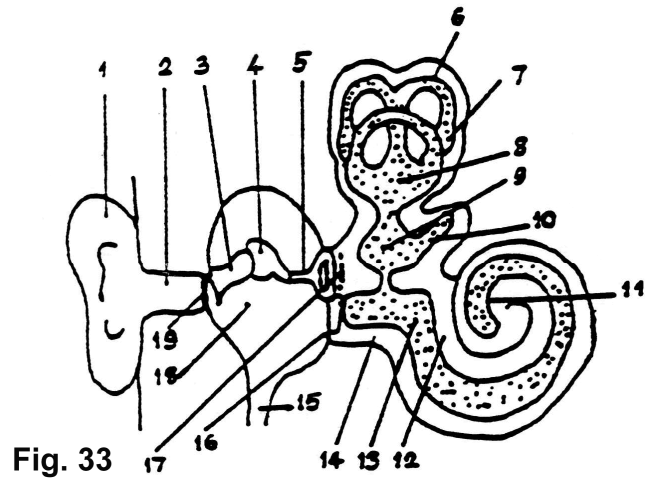


Fig. 33

Fout! Bladwijzer niet gedefinieerd. A. Eustachian tube	nr:
B. malleus (hammer)	nr:
C. semicircular canals	nr:
D. oval windows	nr:

- 132(2pt) Write down the numbers of the following structures as appearing from figure 34:

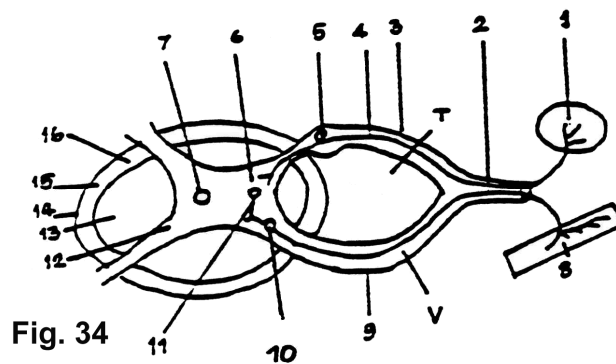


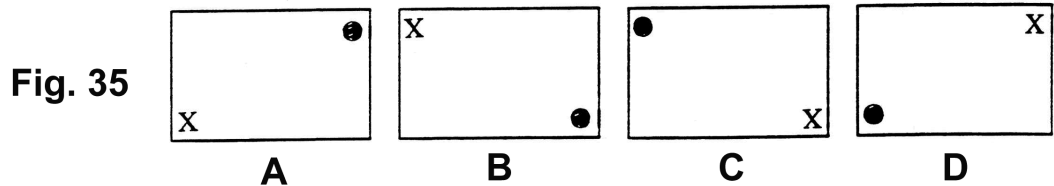
Fig. 34

Fout! Bladwijzer niet gedefinieerd. A.	nr:
B. white matter of spinal cord	nr:
C. spinal ganglion	nr:
D. spinal nerve	nr:

- 133.(2pt) The statements below are partly related to figures 33 and 34. Indicate in the margin with + or — which statements are true (+) or false (—).
- A. Bilateral destructions of parts 11, 12, 13 and 14 of figure 33 lead to deafness.
 - B. Cutting certain spinal nerves which originate from the neck region of the spinal cord results in deafness.
 - C. Cutting the dorsal roots of spinal nerves which originate from the chest and the

- belly regions of the spinal cord results, among others, in the loss of the skin's sensitivity to tactile contact stimuli in certain regions of the body surface.
- D. Eliminating the left hemisphere of the cerebrum results in the loss of the skin's sensitivity (=skin senses) mainly in the left part of the body.

- 134.(2pt) Somebody has a small injury at the retina of the left eye. This injury is situated at the top of this retina, into the direction of the left ear of this person. He is looking with his left eye to a paper with a cross and a spot on it. The image of the spot is exactly projected on the injury of the retina. Choose from figure 35 the correct combination of the positions of spot and cross on the paper.



Answer (combination):

- 135.(2pt) The daily food intake of a 30 year old woman contains 70 g of protein, 70 g of fat and an unknown amount of carbohydrate.
Assume energy values per gram are 17 kJ for protein, 37 kJ for fat and 16 kJ for carbohydrate.

- A. How much is the maximum energy yield of the fat in the diet?

Answer: kJ

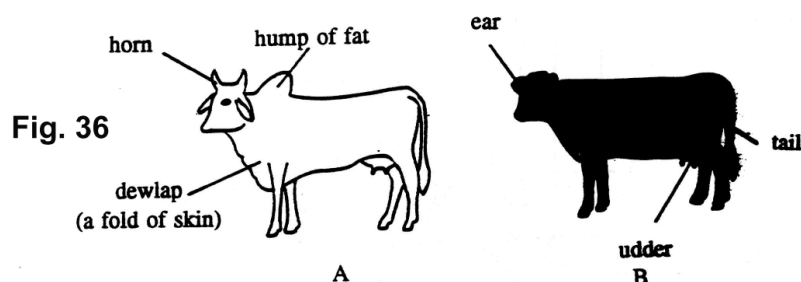
- B. If the energy needs of the woman are 10800 kJ per day, how much carbohydrate must she consume to be in energy balance?

Answer: g

- 136.(2pt) A person eating an entirely vegetarian diet must usually eat a greater mass and variety of protein per day than a person eating a diet containing animal protein. Consider the statements A, B, C and D about this. Indicate in the margin with + or — which statements are a correct (+) or incorrect (—) explanation for the necessary differences in the two diets.

- A. In general, the diversity of amino acids essential for human beings is not as great in a plant protein as it is in an animal protein.
- B. In general, the ratio between the amounts of various essential amino acids in a plant protein differs strongly from that in an animal protein.
- C. In general, the percentage of proteins is lower in vegetable food than in animal food.
- D. In general, plant proteins are more difficult to be digested and less completely digestible than animal proteins.

- 137.(1pt) Figure 36 compares the appearances of two types of cows, A and B.



Which type of cow is better adapted to withstand high temperatures? Fill in A or B.

Answer (Type):.....

138.(2pt) In an experiment in a research station in the tropics both types of cows were moved between sun and shade conditions. Figure 37 shows the effects of these different conditions on body temperature and breathing rate.

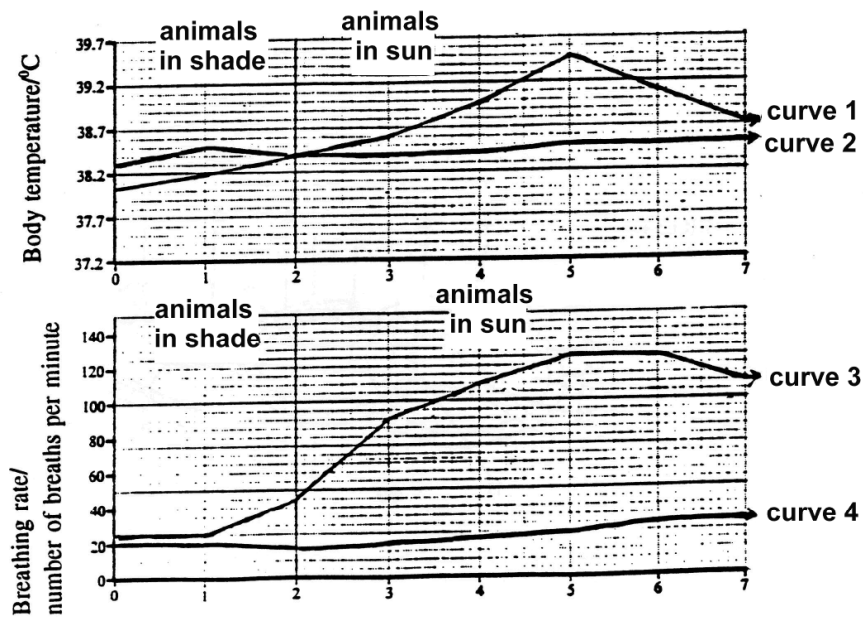


Fig. 37

Indicate for each curve in this Figure which type of cow it represents. Fill in A or B.

Answer curve 1: type
 curve 2: type
 curve 3: type
 curve 4: type

139.(2pt) Match each of the organisms listed below with its corresponding trophic level

- | | |
|------------------|-----------------------|
| A. earthworm | 1. primary producer |
| B. phytoplankton | 2. primary consumer |
| C. pike | 3. secondary consumer |
| D. grasshopper | 4. detritivore |
| E. frog | 5. tertiary consumer |

Answer: A matches with:(Fill in 1, 2, 3, 4 or 5, etc)
 B matches with:
 C matches with:
 D matches with:

E matches with:

140.(2pt) Match each of the terms with the correct definition.

A mutualism	1. camouflage
B commensalism	2. animals with effective defenses are often brightly coloured
C cryptic colouration	3. one partner benefits without significantly affecting the other
D aposematic colouration	4. both partners benefit from the relationship
E mimicry	5. a harmless species resembles an 'armed' one

Answer: A matches with: (Fill in 1, 2, 3, 4 or 5, etc)
 B matches with:
 C matches with:
 D matches with:
 E matches with:

END

Answer key theoretical test

(Part A)

	1	2	3	4	5	6	7	8	9	0
1 - 10	C ²	D ²	C	D	A	D	B	B	C	-
11 - 20	B	D	B ²	C	B	E	C ²	C ²	D	B
21 - 30	B	E	D	D	C	D	A	C	D	A ²
31 - 40	D ²	D	C	D	D	B	A	C	A	D
41 - 50	C	C	B	-	B	B	A	-	B	B
51 - 60	B	C	C	B	B	A	C	C	B	B
61 - 70	-	-	A	D	D	B	D	C	C	A ²
71 - 80	C	B	D	C	A	-	D	-	D	C
81 - 90	C	E ²	C	D	B	B	B ²	C ³	C ³	C
91 - 100	-	A	E	C	B	A	B	D	C	A ²
101 - 110	C	D	B ²	B ²	B ²	E	A	-	B ³	C ²
111	C ²									

Every question has a score of 1 point with a few exceptions;
these are indicated in superscript.

Questions indicated with '-' were skipped.

Total number of points for part A: 124

Answer key Theoretical test (Part B)

112	ssDNA 2,4,6	123	skipped	133A	+
	ds DNA/RNA 1,3,5,7	124A	-	B	-
113A	300	B	+	C	+
B	3200	C	-	D	-
114A	162 163 164 165	D	+	134	B
	CAU AUG GAG UAU	125A	9	135	skipped
	ACA UAU GGA GUA	B	5	136A	+
B	addition	C	10	B	+
C	U	D	7	C	+
115	T B	126A	+	D	-
	N F	B	-	137	A
116	A	C	+	138	B
	B	D	-		A
	C	127A	-		B
117	p - s	B	+		A
	p c -	C	+	139	4
	- - s	D	-		1
	p c -	128A	-		5
	- c -	B	+		2
118	less	C	-		3
	less	D	+	140	4
	less	129	C		3
	less		D		1
	less		E		2
	same		B		5
119	skipped		A		
120	true	130A	-		
	true	B	-		
	false	C	+		
	false	D	+		
121A	3	131A	15		
B	3	B	3		
C	5	C	6, 7		
D	5	D	17		
122A	4, 5, 6	132A	10		
B	5, 6, 7	B	13		
C	(4), 5, 6	C	5		
D	4, 5, 6, 7	D	2		

Total number of points for part B: 63